Brain MRI Tumor Segmentation and Quantitative Analysis Using Classical Image Processing

Week 9 - Completion and Improvements

Kush Kelaiya
Enrollment No.: AU2440018
School of Engineering and Applied Science
Ahmedabad University
Ahmedabad, Gujarat, India

Rushin Modi
Enrollment No.: AU2440028
School of Engineering and Applied Science
Ahmedabad University
Ahmedabad, Gujarat, India

Kritee Shah
Enrollment No.: AU2440080
School of Engineering and Applied Science
Ahmedabad University
Ahmedabad, Gujarat, India

Jash Odedara
Enrollment No.: AU2440096
School of Engineering and Applied Science
Ahmedabad University
Ahmedabad, Gujarat, India

Submitted By:- G9 - The Medics.

I. Work done this Week:-

We have completed the base detection and segmentation of the brain MRI scans. This week, our primary focus has been to improve the accuracy and reliability of the system. Since we have been instructed to perform the entire process using classical image processing methods only, achieving accuracy comparable to industry-standard methods has been a challenge. However, we are continuously trying to achieve the maximum possible accuracy within the limitations of classical image processing techniques.

We have gone through multiple iterations by modifying and experimenting with different formulas to improve the detection of the tumor region. During this process, we observed that several accuracy calculation formulas were producing false values, which affected the reliability of the results. Therefore, we had to compare different approaches and examine which method was providing more meaningful results for our dataset. After these iterations, we decided to use subtraction with Absolute values to determine the regions where the tumor is present. This approach allows us to identify the differences between the scans and compare the detected regions with the .json values provided in the dataset.

We have also explored different masking and filtering methods to ensure that the skeletal region, surrounding brain matter, and other regions of the human body do not interfere with tumor detection. Our main objective was to prevent unwanted regions from affecting the segmentation process and to ensure that the tumor region could be identified without being blocked by other structures. We have primarily worked on this through skull stripping and other masking processes, including Parenchymal Band Thresholding and Central Intracranial Connected Components. Along with these techniques, we have experimented with other similar approaches to work with the brain matter and distinguish the relevant regions from other structures present in the head region.

However, when we tried to apply the same method to the base scan that we are currently using, the system was able to detect the tumor region but was also producing negative accuracy values. This indicates that the current accuracy calculation is not providing consistent results for the base scan. We are still working on understanding and correcting this behaviour accordingly. When we consider the other scans throughout the dataset, the results are comparatively better. The accuracy method we have used is Bounded Volumetric Ratio + Hemisphere-Gated Spatial Decay. Using this method, we have been partially successful in detecting the tumor regions across most of the scans in our dataset.

After evaluating the results obtained from the current implementation, the overall accuracy value throughout the images where we have to detect the tumor is approximately 41.1% (rounded off). However, when considering only the scans that actually contain a tumor, the accuracy of tumor detection is approximately 67.4% (rounded off). These results represent the current performance of our system, and we are continuing to work on improving the detection accuracy while remaining within the constraints of classical image processing methods.

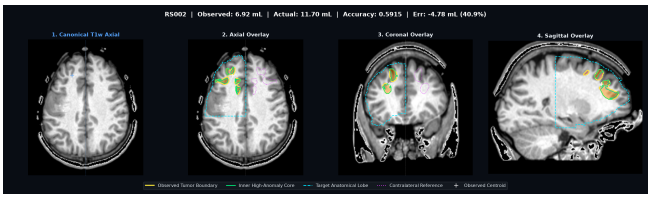


Fig. 1. False Result

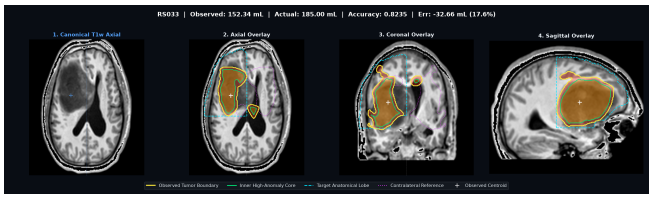


Fig. 2. True Result

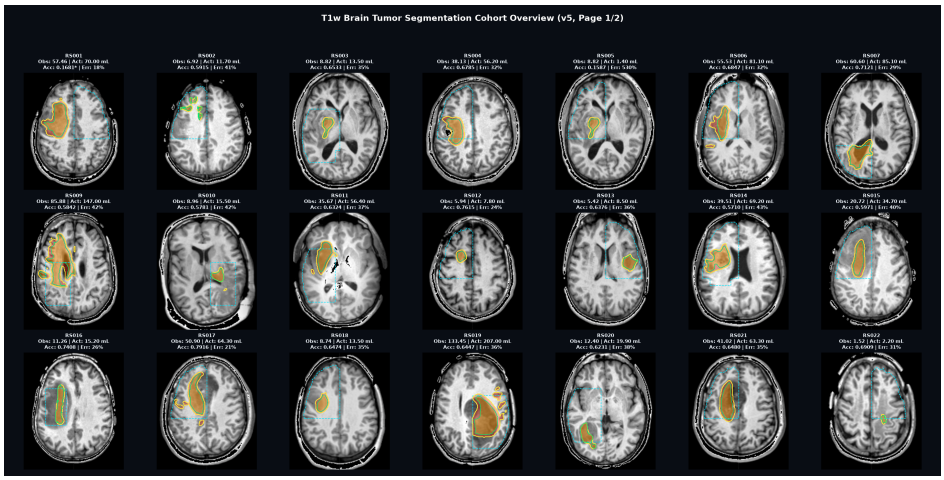


Fig. 3. Results Pg1

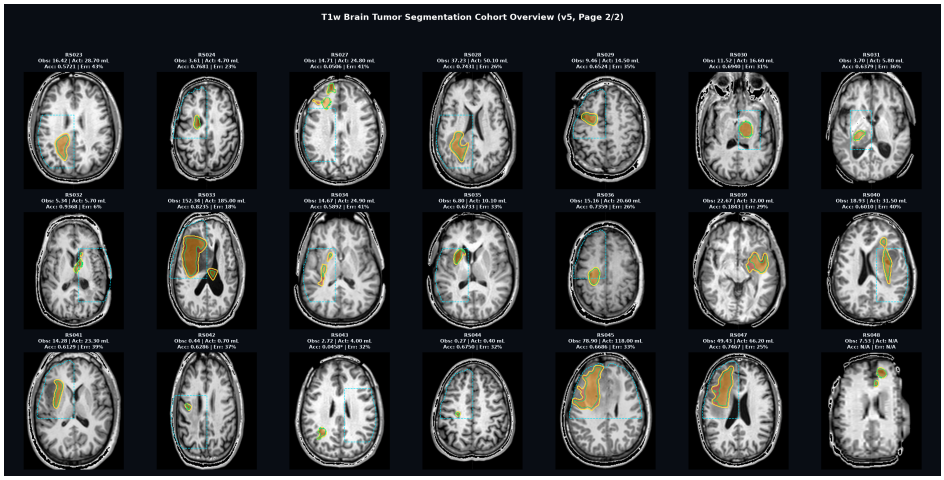


Fig. 4. Results Pg2

Repo Link:-

https://github.com/Rushin-Mod/DIP_2026_P20_G9